

# Zhengyang Kris Weng

Robotic Systems R&D - Surgical/Humanoid Robotics

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## Professional Summary

Robotics engineering tech lead with 5+ years developing surgical and domestic robotic systems at leading companies. Expert in robotic arm and actuator design, calibration systems, and control software with proven track record in system integration and R&D. M.S. in Robotics with focus on dexterous manipulation, robot control and machine learning. Strong background in mechanical design, embedded systems, and real-time control.

## Professional Experience

### Robotic Systems Engineer

*Sunday Robotics, Mountain View, CA*

Feb 2026 – Present

- **System Engineering:** Evaluate, integrate and debug robotic subsystems; develop and maintain motor and sensor drivers (C++/ Python); design and develop robotic calibration and bring-up processes.
- **Scaling Policy Evaluation:** Develop infrastructure and agents to support and triage autonomous fleet operation and perform large-scale trained policy rollout and evaluation.

### Senior Mechanical Engineer

*Johnson & Johnson MedTech, Santa Clara, CA / Remote*

Oct 2022 – Feb 2026

- **System Hardware R&D:** Core team member for v2 robotic arm development, contributing to system specifications, design reviews, and validation strategy for the Monarch robot surgery platform
- **Verification & Validation:** Developed test cases and fixtures and conducted V&V for surgical robotic devices, ensuring compliance with performance and safety requirements
- **Software Development:** Developed robot calibration software in C++ and designed admittance control visualization prototypes using Python and CoppeliaSim
- **Innovation & IP:** Invented fluid management system for Monarch Urology procedures (WO2025257720A1)
- **Leadership & Development:** Transitioned to remote part-time work while pursuing M.S. in Robotics (2024-2025), maintaining full project responsibilities and deliverables

### Mechanical/Robotics Engineer

*Neocis Inc., Miami, FL*

Jun 2021 – Oct 2022

- **Mechatronics R&D:** Designed a 7-DoF robotic guidance arm for the Yomi S robot, including kinematic analysis and component specification. Developed a series of custom compact high-precision joint actuators from scratch
- **System Integration:** Led build and integration across three prototype arm iterations, achieving sub-millimeter accuracy for robotic dental surgery
- **Calibration Algorithms:** Developed optical robot kinematic calibration methods using optimization with regularization, enhancing accuracy and robustness
- **Leadership:** Mentored interns and delivered training to new hires on robotics hardware team
- **Career Progression:** Promoted to Senior Engineer (Aug 2022) in recognition of leadership and contribution

### Mechanical Engineer Co-op

*Harmonic Bionics Inc., Austin, TX*

May 2020 – Dec 2020

- **Exoskeleton Development:** Designed robotic systems for 14-DoF rehabilitative upper extremity exoskeleton, including linear sizing mechatronic systems and handheld user interface
- **Operations:** Launched company machine shop, authored safety SOPs, and manufacturing equipment training

## Education

### M.S. in Robotics

*Northwestern University, Evanston, IL Focus: Dexterous Manipulation, Kinematic Control, Imitation Learning*

Sep 2024 – Aug 2025

### B.S. in Mechanical Engineering

*Georgia Institute of Technology, Atlanta, GA Concentration: Robotics and Control Systems*

Sep 2016 – May 2021

## Open-Source Projects

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**Hand Tracking Streamer: Meta Quest VR App for Tracking Hand Landmark Telemetry** 2026  
*Zhengyang Kris Weng, San Jose, CA*

- Developed a lightweight hand telemetry utility and companion SDKs that turn a Meta Quest headset into a precision controller for robotics teleoperation and motion capture. Available on Meta Horizon Store for free

**LeVR: A Modular VR Teleoperation Framework for Imitation Learning in Dexterous Manipulation** 2025  
*Northwestern University, Evanston, IL*

- Developed modular VR teleoperation framework integrated with LeRobot platform for imitation learning in robotic manipulation
- Released open-source implementation (LeFranX) for Franka robotic arms and RobotEra XHand manipulator with high-frequency teleoperation and imitation learning
- Collected dataset of 100+ expert demonstrations and validated framework with state-of-the-art visuomotor policies

**BiDexHand: Open-Source 16-DoF Dexterous Hand** 2024-2025  
*Northwestern University, Evanston, IL*

- Designed low-cost 3D-printed 16-DoF anthropomorphic hand with a complete ROS 2 control stack for joint-level control, functional retargeting, inverse kinematics, simulation and teleoperation functions
- Implemented vision-based calibration using AprilTags and integrated numerical IK solver with MoveIt!2 for real-time control with Franka robot arm

## Technical Skills

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|----------------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>Robotics &amp; Simulation</b> | Robot kinematics & dynamics, kinematic calibration, ROS 2, CoppeliaSim, Gazebo, MuJoCo, PyBullet, RViz, Unity |
| <b>Software Development</b>      | Python (NumPy, Pandas, OpenCV), C++, MATLAB, Git, Linux, Bash                                                 |
| <b>ML Frameworks</b>             | PyTorch, TensorFlow, HF Transformers, scikit-learn                                                            |
| <b>Mechanical Design</b>         | SolidWorks (CSWE), OnShape, Fusion 360, ANSYS FEA, rapid prototyping, 3D printing (SLA, FDM, SLS, MJF)        |
| <b>Electrical</b>                | PCB design (KiCAD), oscilloscope, soldering                                                                   |
| <b>Machining</b>                 | Milling, lathing, water jetting, laser cutting                                                                |
| <b>Languages</b>                 | English (native), Mandarin Chinese (native)                                                                   |

## Publications & Patents

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- **Z.K. Weng**, M. Elwin, H. Liu, "LeVR: A Modular VR Teleoperation Framework for Imitation Learning in Dexterous Manipulation", *arXiv preprint*
  - **Z.K. Weng**, "BiDexHand: Design and Evaluation of an Open-Source 16-DoF Biomimetic Dexterous Hand", *IEEE ICRA 2025 Dexterity Workshop*, Spotlight Presentation
  - A. Harapanahalli\*, E. Muly\*, H. Welch\*, T. Brumfiel\*, **Z.K. Weng\***, et al. "Towards a Biomimetic and Dexterous Robot Avatar: Design, Control, and Kinematics Considerations," *IEEE/ASME AIM 2020*, Best Breaking Results Poster (\*equal contribution)
  - International Patent WO2025257720A1 — "Fluid Management System for Medical Procedures"
  - US Patent US11842304B2 — "Accessible Ride Hailing and Transit Platform"

## Awards & Recognition

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- IEEE ICRA 2025 Dexterity Workshop — Spotlight Presentation for BiDexHand research
  - IEEE AIM 2020 — Best Late Breaking Results Poster for biomimetic robotics
  - First Place, 2021 Georgia Tech VIP Innovation Competition (Hardware, Devices & Robotics)
  - Georgia Tech President's Undergraduate Research Award (2021)
  - Toyota Mobility Foundation Challenge Winner (2017)
  - Certified SolidWorks Expert (CSWE) — Highest mechanical design certification awarded by Dassault Systèmes